

Rachel Atkeison

MUSIC TECHNOLOGY / AUDIO SOFTWARE ENGINEERING INTERN

Location Atlanta, GA, United States • **Mobility** Open to relocation to Japan

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Portfolio rachelatkeison.github.io • **GitHub** github.com/rachelatkeison • **LinkedIn** linkedin.com/in/rachel-atkeison

SUMMARY

Computer Science student focused on building real-time music software, digital instruments, and immersive audio systems. Experienced in C++ audio development, JUCE, DSP, MIDI processing, and interactive music tools through hands-on projects in synthesizer design, musical intelligence, and spatial performance environments. Seeking an opportunity to learn, contribute, and continue growing as an engineer while deepening my experience in audio software, music technology, digital instruments, and immersive sound systems.

EDUCATION

University: Kennesaw State University

Degree: Bachelor of Science in Computer Science

Graduation: Expected 2027

Academic standing: GPA 4.0 • Dean's List

Relevant coursework: Digital Signal Processing, Computer Graphics, Systems Design, Software Engineering, Advanced Programming

TECHNICAL PROFILE

Programming: C++, Python, Java, C#

Audio stack: JUCE, VST3/AU plugin development, DSP, MIDI systems, synthesis, real-time audio processing

Frameworks / tools: Unity, FMOD, OpenGL, OSC, Git

Systems: Low-latency audio pipelines, multithreaded systems, interactive audio software

Languages: English (fluent), Japanese (conversational)

SELECTED PROJECTS

LumenBloom

Polyphonic Synthesizer Plugin | C++, JUCE, DSP

- Built a polyphonic software synthesizer for expressive digital instrument design and real-time audio processing.
- Implemented oscillator synthesis, ADSR envelopes, filtering, modulation, and stereo shaping for flexible sound design.
- Created musician-facing parameter controls and visual feedback to support an interactive plugin workflow.

Aureine Music Box

Real-Time MIDI Analysis System | Python, PySide6, PyQtGraph, MIDI

- Developed a real-time MIDI analysis tool that interprets live musical input during performance.
- Implemented chord detection, key and scale recognition, interval tracking, and harmonic analysis using music-theory logic.
- Designed an interactive desktop interface for visualizing musical structure in real time.

CloudStage MR Studio

Spatial Performance Environment | Unity, FMOD, Spatial Audio

- Created an immersive virtual performance environment for spatial music experimentation.
- Implemented dynamic sound positioning, staged audio interaction, and responsive behavior for virtual musical experiences.
- Explored how software, space, and sound can function together as a performance instrument.

EXPERIENCE & CREATIVE PRACTICE

Computer Science Tutor / English Tutor — Kennesaw State University | Dec 2021 - May 2025

- Provided individualized instruction in programming, writing, and problem-solving to undergraduate students.
- Adapted technical explanations to different learning styles, strengthening communication and teaching ability.

First Frost — Original Lo-fi Music Project

- Produced and released an original lo-fi music project centered on piano texture, atmosphere, and sound design.
- Personal music creation directly informs how I think about musician-facing software and digital instrument design.